— Solution notes —

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

9 Further Human–Computer Interaction (afb21)

Designing smart systems

(*a*) Briefly explain, including quantitative estimates, how the information content of English text compares to the information rate that could theoretically be communicated using a computer keyboard. [2 marks]

Answer: The average entropy of English text is relatively low, because of standardised word spelling and conventional phrasing, somewhere around 1 bit per character. The standard computer keyboard has around 60 keys, so user selection of any one of those keys could potentially communicate 7 bits of information.

Designing efficient systems

(*b*) How could the relationship you have described be used in a design to enter text either more quickly, or with less effort from the user? [2 marks]

Answer: Predictive text can enter text more quickly by mapping longer strings of characters to fewer keypresses (perhaps using soft keys with visual labels or mnemonics). Alternatively, lower information rates might allow users to select probable texts more slowly but with less concentration, for example using the single-finger operation as supported by the Dasher interface.

Bayesian approaches to interaction

(*c*) Consider an operating system command line interpreter that interprets text commands on a probabilistic basis. Explain how each of the elements of Bayes' theorem (Posterior, Likelihood, Prior and Evidence) would be applied in the implementation of this interpreter. [8 marks]

Answer: Prior: what was the probability that the user was going to type this command (perhaps based on recent command history)

Likelihood: how likely is this command in general (perhaps based on a corpus of commands executed by other users of the same system)

Evidence: how closely does the sequence of characters that the user just typed match the command (for example, are there any spelling errors)

Posterior: does it seem likely that this is what the user wants to do? If the posterior probability is low, the user could be asked for confirmation, or be prompted with additional information.

Biases and heuristics

(*d*) Give one example of a well-known bias or heuristic that might be applied by users of a command line interpreter when they need to make rapid decisions. Explain, using Bayes theorem and/or information theory, what could make that strategy effective in the above scenario. [4 marks]

Answer: The availability heuristic refers to decisions made on that basis that something comes to mind easily, rather than having thought through the alternatives. For example, the command "rm -rf"' is a traditional cliche for Unix users. I might type the command just because I can remember it easily, without stopping to think whether the "f" option is really required or not. The availability heuristic can be considered as using the associative memory capabilities of the human brain to provide a Bayesian prior on the basis of previous experience.

Attention investment

(*e*) Explain how the attention investment model of abstraction use might relate to

the use of more or less abstract strategies in the scenario you have described.
[4 marks]

Answer: Using the conventional and memorable command has low attention costs, but also brings risks that additional attention might be required in future, in order to fix a costly mistake that resulted from not having carefully thought through the consequences. In principle, I could modify my shell configuration script to define an alias for a memorable and less risky deletion command to be used in future. However, this potential cost saving and risk reduction has up-front costs in formulating and expressing the abstraction that I need, which will require imagining the potential circumstances where it is going to be applied in future. The decision to invest attention in modifying my shell is therefore based on my estimate of the number of times I might ever use that command in future.
